Donegal House Price Index

Release rel-57ed44641cde · published 09 October 2026 · register data to 02 October 2026

What this is. A house price index for County Donegal built from every home that sold more than once in the public Property Price Register, plus a lookup that brings any Donegal home's past sale price forward to today's market, with an honest range. Every figure in this release has passed the project's quality gates.

242 Donegal index 2025 (2015 = 100)	**2.42×** price change 2015 → 2025	**98.0%** property links correct (260 human labels)	**25.6%** typical estimate error vs 37.8% unadjusted
21,484 register rows in the release	**1,185** repeat-sale pairs behind the index	**90%** prediction range tested on later sales	**published** release state (all gates passed)

Scope	County Donegal, with a Letterkenny town index (decision D-030)
Product	Lookup app (Databricks Apps) and AI/BI dashboard; this report
Refresh	Monthly, 10th at 06:00 Europe/Dublin; each refresh is a new release
Author	Conor McCool

1. The release in sixty seconds

The release, step by step:

- **The problem.** Ireland publishes every home sale since 2010, but with no ID for the home itself: the same house appears under differently typed addresses. You cannot simply ask how much a given home, or a given town, has changed.
- **The data.** I took the official register untouched into Databricks: 21,484 Donegal sales, every row accounted for, with VAT and non-market sales handled by written, tested rules.
- **Linking sales to homes.** A probabilistic record-linkage model (Splink) with safety rules decides which sales are the same home. I labelled 260 pairs by hand to measure it: **98% of links are correct**, and it finds about 75% of true repeat sales.
- **The index.** Comparing each home with itself (a Case-Shiller weighted repeat-sales index) shows Donegal prices **2.4× their 2015 level by 2025**. A bootstrap gives each year a confidence band, and years without enough evidence are suppressed, not guessed.
- **Checked against official figures.** Year-on-year growth tracks the CSO's regional index with almost no bias (+0.3 points a year).
- **Tested like a forecast.** Each later sale was predicted from an index that had never seen that home or any later data. The typical error was **26%** against 38% for the old price unadjusted, and the 90% ranges held up on the test years.
- **The product.** A signed-in lookup app: type an address, see the home's sale history, and get its last price brought forward, with a range and plain reasons when no fair number exists. A dashboard shows the market view.
- **Built to be trusted.** Each release is immutable and reproducible (identical on my laptop and on Databricks), refreshed monthly, and can be rolled back.

> **One line:** *a Databricks-built, quality-gated Donegal house price index and lookup that tells people what their home's last sale price means in today's market, and how sure we can be.*

2. Who it is for and how they use it

User	Their question	What they do	What they get
Homeowner	How has my home moved since I bought it?	Search the address; anchor on the purchase; pick the latest year	Purchase price brought forward, with a 90% range
Buyer	Is the asking price in line with the market since its last sale?	Anchor on the last sale; compare the estimate with the asking price	A market-only benchmark; a big gap is a prompt to ask why
Seller	What would the market alone suggest?	Same as the buyer	A starting point for a conversation with an agent
Local reader	How have Donegal and Letterkenny prices moved?	Open the dashboard	Index, uncertainty, CSO comparison, data quality

What it is not. Not a valuation, a mortgage or insurance figure, or a forecast. It cannot see extensions, renovation, condition or the particular street, which is why the range for one home runs from roughly half to double the estimate.

Getting access

- The app and dashboard run on Databricks and need a sign-in: this is an authenticated release, with no anonymous public access.
- A workspace admin invites the user, then grants **Can use** on the app `ireland-rsi-lookup-donegal` and view access to the dashboard.
- People without an account get this report, which carries the published index, worked examples and the evidence.

Using the lookup

Step	What the user sees
1. Type an address	Candidate homes; similar-looking addresses are never merged automatically
2. Pick the home	Every recorded sale: date, price as filed, VAT-adjusted price where relevant, and whether it can anchor
3. Choose anchor and target	Defaults: most recent usable sale and the latest published year
4. Read the result	'Index-adjusted estimate as of [year]', the 90% range, the index change, which index was used, provisional flags
5. Check the chart	The Donegal index with its own (narrower) confidence band

When there is no estimate

Reason shown	Meaning
No eligible anchor	Every sale is non-market, a bulk filing or VAT-unresolved; the facts are still shown
Unsupported period	The sale or target year is suppressed for lack of evidence
Sparse geography	No published index covers the area
Uncalibrated interval	The holding period failed its accuracy test (none currently)

3. Worked examples from this release

Real register records, run through the same service the app uses. The later resales illustrate the range; they are not held-out tests, because they are part of this index. The held-out results are in section 5.

Home (register address)	Anchor sale	Estimate for 2025	90% range	What happened
26 Maginn Avenue, Buncrana, Co. Donegal	Feb 2016 · €80,000	**€176,660** (×2.21)	€74,838 to €320,697	Resold Dec 2024 for €270,000: inside the range, above the estimate
24 Donegal Road, Ballybofey, Co Donegal	Dec 2014 · €102,000	**€295,134** (×2.89)	€125,026 to €535,766	Resold Apr 2025 for €252,250: inside the range, below the estimate
No 10B Fairgreen Park, Letterkenny, Co Donegal	Apr 2016 · €60,000	**€132,495** (×2.21)	€56,128 to €240,523	Letterkenny home; the Donegal index is used (D-025)

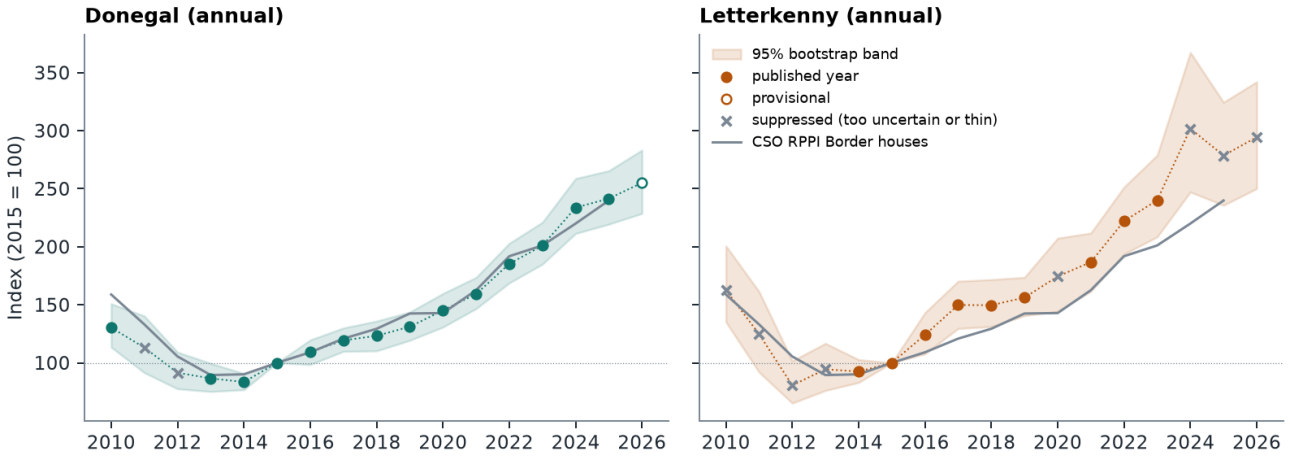
When the app refuses

Request	Result
29 Amalfi Court, Letterkenny, anchored on the April 2019 sale	**NO_ELIGIBLE_ANCHOR**: This property has no sale that can anchor an estimate. The register marks that sale as not at full market price, so the app refuses it rather than switching anchors.
21 Carraig Craobh, Crievesmith, Letterkenny, anchored on its December 2012 sale, target 2025	**UNSUPPORTED_PERIOD**: No published index series covers both the sale period and the target period. 2012 is suppressed in the Donegal index (too few resales to measure it reliably), so no fair number exists for that anchor.

Reading the Buncrana example aloud: *'It last sold for €80,000 in 2016. Donegal prices are 2.2 times higher now, so the market alone takes it to about €176,000, and nine times in ten a home like this sells somewhere between €75,000 and €320,000. It actually sold for €270,000, which tells you the house or its street did better than the county average.'*

4. The published index

Published release rel-57ed44641cde: annual repeat-sales indices



Annual repeat-sales index, 2015 = 100, with 95% property-history bootstrap bands, from this release. Crosses mark suppressed years; the grey line is the CSO RPPI for Border houses.

Year	Donegal	95% band	Pairs	Letterkenny	95% band
2010	130.3	113-151	37	suppressed	
2011	suppressed		33	suppressed	
2012	suppressed		59	suppressed	
2013	86.6	75-100	53	suppressed	
2014	83.5	77-91	155	92.5	83-103
2015	100.0	100-100	174	100.0	100-100
2016	109.4	98-120	172	124.1	108-143
2017	119.4	110-130	139	149.8	130-170
2018	123.3	110-136	183	149.7	131-172
2019	131.1	119-144	153	156.2	141-174
2020	145.1	131-160	124	suppressed	
2021	159.3	147-174	203	186.8	162-212
2022	185.3	169-203	220	222.1	194-251
2023	201.0	185-221	184	240.1	208-279
2024	233.7	211-259	178	suppressed	
2025	241.5	219-265	178	suppressed	
2026	255.1 (prov.)	229-283	125	suppressed	

A year is published only with at least 20 repeat-sale pairs and a 95% band narrower than about ±15%. 2011-2012 fail for Donegal. Most Letterkenny years after 2023 fail because too few homes resold to measure them reliably. Quarterly series are calculated but not published: they fail the bootstrap gate.

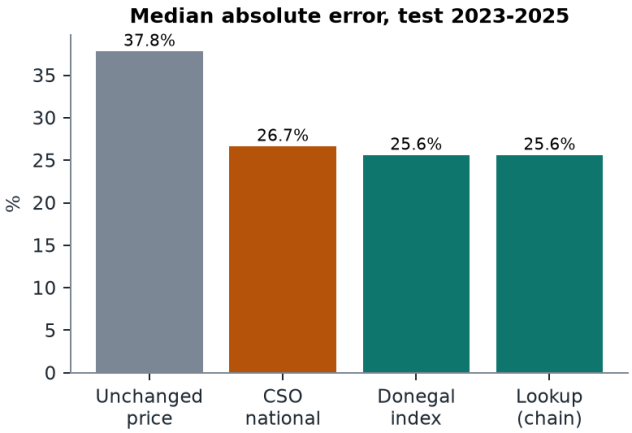
5. How far to trust it

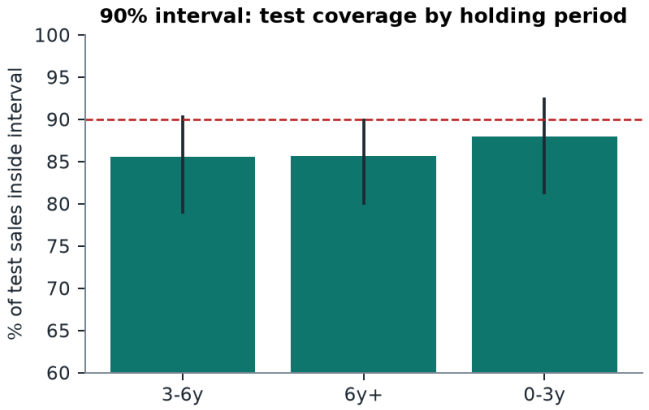
Question	Evidence in this release
Is every source row accounted for?	21,484 rows, each with one disposition; the original file is kept byte-for-byte
Are sales linked to the right homes?	98.0% of links correct (95% CI 95.6%-99.7%); recall 75%; 260 labels (71 same, 122 different, 67 can't tell)
Does the index agree with official figures?	Year-on-year growth against CSO Border houses: bias +0.3 pp, mean absolute difference 5.7 pp (12 published years)
Do the modelling choices change the answer?	Link thresholds from 0.5 to 0.97 move 2015-2025 growth by ≤ 0.04 log points; the estimators agree within 0.015
Are estimates better than doing nothing?	Typical error 25.6% vs 37.8% unadjusted and 26.7% using the CSO national index (446 test sales, 2023-2025)
Can it be reproduced and rolled back?	Identical release ID locally and on Databricks; immutable releases; rollback drills passed

Property prediction ranges (90%)

Horizon	Calibrated range (× estimate)	Test coverage	Test sales
0-3y	0.56 to 2.32	88.0%	125
3-6y	0.62 to 1.92	85.6%	139
6y+	0.42 to 1.82	85.7%	182

Coverage is slightly under 90% in every band but within the 5-point tolerance. The ranges are wide because they include everything the index cannot see about an individual home.





6. Scope and limitations

- **Donegal only (D-030).** This is a complete county release. The national pipeline exists but is not published: each county would need its own labelled evidence first.
- **Resold, identifiable homes.** Rural homes are 56% of Donegal sales but 38% of the repeat sales the index uses, because townland-only addresses cannot be linked safely. The index leans urban and lower-priced.
- **About 2% of links join two different homes**, mostly generic apartment addresses in different towns. That is short of the 99% aspiration; a fix is proposed for testing on fresh labels.
- **Retrospective testing.** Accuracy was measured on past data. The system has no archive of what was known month by month, so it makes no real-time accuracy claim.
- **No houses/apartments split** exists in the register, so the CSO comparison is approximate.

7. Release facts

Release	rel-57ed44641cde (published), created 2026-10-09 16:33
Source snapshot	ppr:county=Donegal:5729e360427cc33d (PSRA register, data to 02 Oct 2026)
Versions	addresses addr_v1+ref-d81f89a66c28+55a84bf2, backtest bv-7eb627e7f022, clean qr-2026-10-06+20fcac555c, entities ent-dbf7452a48d5, index fv-3a7219ee3358, linkage_production lm-31cc8abb732d, uncertainty uv-71a8f2c5da37, validation vv-94dd8f3d284f
Gates	all_stages_present ✓, g10_validation_recorded ✓, g11_calibrated_bands ✓, g6_threshold_frozen ✓, g7_pairs_audited ✓, g9_supported_series ✓
Databricks runs	full refresh (this release): 358191034653175
Reproduce	python jobs/run_pipeline.py --profile donegal --start clean · databricks bundle run ingest_ppr -t donegal
Documents	docs/user_guide.md, methodology.md, validation_report.md, backtest_report.md, runbook.md, decisions.md (D-001…D-030)